

Home Network Vulnerability Assessment Using Kali Linux and Nmap

1. Executive Summary

This assessment evaluates the security posture of the local home network. The objective was to identify exposed services, detect potential vulnerabilities, and assess risks associated with network-accessible devices.

Testing was conducted using a Kali Linux virtual machine configured within the internal network. Initial findings indicate that the network gateway exposes web management services over HTTP and HTTPS while other potentially sensitive services are filtered by the firewall.

No critical vulnerabilities were identified during this assessment, though the router management interface remains a potential attack surface if not properly secured.

2. Scope of Assessment

Category	Description
Assessment Type	Internal Network Vulnerability Assessment
Target Network	192.168.0.0/24
Primary Target	Router (192.168.0.1)
Testing Platform	Kali Linux Vm
Tools Used	Nmap

Limitation

- Testing limited to devices within the local network
- No exploitation activities performed
- Only reconnaissance and vulnerability identification conducted

3. Methodology

The vulnerability assessment followed a structured methodology consisting of the following phases:

- Environment Setup
- Network Discovery
- Port Scanning
- Service Enumeration
- Vulnerability Identification
- Risk Analysis

4. Environment Setup

The assessment environment was prepared using a Kali Linux virtual machine running on **Oracle VirtualBox**.

The virtual machine network adapter was configured in **Bridged Mode**, allowing the Kali system to function as a device on the same network as other local devices.

Network configuration verification was performed using:

Ip a

Result:

Interface: eth0

IP Address: 192.168.0.27

Network: 192.168.0.0/24

5. Network Discovery

Command used:

nmap -sn 192.168.0.0/24

purpose:

- 1) To identify active devices
- 2) build asset inventory

Result:

The gateway device **192.168.0.1** was identified as the primary network device.

6. Port Scanning

Command: `nmap 192.168.0.1`

Result:

Port	Service	Status
80	HTTP	Open
443	HTTPS	Open
21	FTP	Filtered
22	SSH	Filtered
23	Telnet	Filtered

7. Service and OS Detection

Command: `nmap -sV -O 192.168.0.1`

Category	Result
Device Type	Wireless Access Point
Operating System	Embedded Linux
Kernel Version	Linux 3.x – 4.x
Vendor	D-Link

Interpretation:

The router appears to be running embedded Linux firmware typical for consumer networking devices.

8. Vulnerability Scanning

Command used: `nmap --script vuln 192.168.0.1`

Purpose: To Identify known vulnerabilities using Nmap NSE scripts.

Results:

Vulnerability Check Result

Avahi DoS	Not Vulnerable
DOM-based XSS	Not Detected
Stored XSS	Not Detected
CSRF	Possible login form exposure

Interpretation:

No critical vulnerabilities were detected during the scan. However, the router login form may lack CSRF protection, which could potentially allow malicious requests if an authenticated user is tricked into visiting a crafted webpage.

Risk Level: Low

9. Network Service Discovery

Command used: `nmap -sV 192.168.0.0/24`

Purpose: To identify exposed services across all devices within the network.

Results:

Router (192.168.0.1)

Port	Service
80	HTTP
443	HTTPS

Windows Device (192.168.0.24)

Port	Service
3580	LabVIEW HTTP Service
5357	Microsoft HTTPAPI
8080	Embedded HTTP Server

Service detection identified the operating system as **Windows**.

The presence of **NI LabVIEW** indicates engineering or development software installed on the host.

Kali Assessment Machine (192.168.0.27)

All scanned ports were closed, indicating the system is not exposing network services.

10. Risk Analysis

Finding	Risk Level	Description
Router Web Management Interface	Low	Router admin panel accessible via HTTP/HTTPS
Possible CSRF Exposure	Low	Router login form may lack CSRF protection
Additional Web Services on Windows Host	Informational	Multiple HTTP services detected

12. Recommendations

To improve the security posture of the network, the following actions are recommended:

- Ensure router firmware is regularly updated
- Disable remote router administration if enabled
- Use strong administrator passwords
- Restrict router management access to trusted devices
- Review services running on internal hosts such as the Windows device exposing HTTP services

13. Evidence

Example commands used during the assessment:

```
ip a
nmap -sn 192.168.0.0/24
nmap 192.168.0.1
nmap -sV -O 192.168.0.1
```

nmap --script vuln 192.168.0.1

nmap -sV 192.168.0.0/24

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a5:b4:bf brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.27/24 brd 192.168.0.255 scope global dynamic noprefixroute eth0
        valid_lft 14190sec preferred_lft 14190sec
    inet6 fc01::a061:cb25:9371:5672/64 scope global dynamic noprefixroute
        valid_lft 2591790sec preferred_lft 604590sec
    inet6 2001:e68:544d:9839:677b:c08c:515b:949d/64 scope global dynamic noprefixroute
        valid_lft 258990sec preferred_lft 172590sec
    inet6 fe80::6c77:ca52:9fdb:f2e3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
$ nmap 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 00:43 EDT
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.011s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
53/tcp    filtered domain
80/tcp    open  http
443/tcp   open  https
515/tcp    filtered printer
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)

Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds

(kali@kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ nmap --script vuln 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 01:01 EDT
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.012s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    open      http
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
| http-csrf:
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=RTK_GW.realtek
|   Found the following possible CSRF vulnerabilities:
|
|     Path: http://RTK_GW.realtek:80/
|     Form id: username
|_   Form action: /boaform/admin/formLogin
443/tcp   open      https
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
|_http-dombased-xss: Couldn't find any DOM based XSS.
515/tcp   filtered  printer
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)

Nmap done: 1 IP address (1 host up) scanned in 65.92 seconds

(kali㉿kali)-[~]
$
```

```

(kali@kali)-[~]
$ nmap -sV -O 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 00:49 EDT
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.20% done; ETC: 00:49 (0:00:02 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 52.55% done; ETC: 00:49 (0:00:01 remaining)
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.0045s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
53/tcp    filtered domain
80/tcp    open  tcpwrapped
443/tcp   open  ssl/https?
515/tcp   filtered printer
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)
Device type: general purpose|WAP
Running: Linux 3.X|4.X, Asus embedded
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4 cpe:/h:asus:rt-ac66u
OS details: Linux 3.10 - 4.11, Asus RT-AC66U WAP
Network Distance: 1 hop

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.09 seconds

```

```

(kali@kali)-[~]
$

```

```

(kali@kali)-[~]
$ nmap -sV 192.168.0.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 02:07 EDT
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.0042s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
53/tcp    filtered domain
80/tcp    open  tcpwrapped
443/tcp   open  ssl/https?
515/tcp   filtered printer
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)

Nmap scan report for HBANBAMBABRK.realtek (192.168.0.24)
Host is up (0.00056s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
3580/tcp  open  http      National Instruments LabVIEW service locator httpd 1.0.0
5357/tcp  open  http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
8080/tcp  open  http      Embedthis HTTP lib httpd
MAC Address: 4C:49:6C:97:66:D4 (Intel Corporate)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for kali.realtek (192.168.0.27)
Host is up (0.000010s latency).
All 1000 scanned ports on kali.realtek (192.168.0.27) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 18.44 seconds

```

```

(kali@kali)-[~]
$

```


14. Conclusion

This assessment provided a baseline evaluation of the local network security posture. The network gateway exposes limited services and appears to have firewall protections in place.

No critical vulnerabilities were identified during the assessment. However, web management interfaces and additional services on internal hosts indicate that proper configuration management and regular firmware updates remain important to maintain network security.